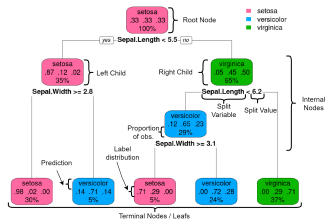
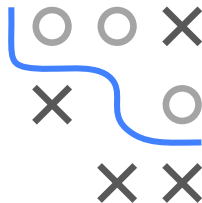


Introduction to Machine Learning

Advanced Risk Minimization

Loss functions and tree splitting

(Deep-Dive)



Learning goals

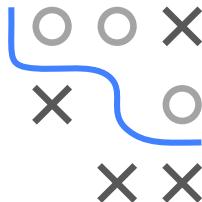
- Tree splitting loss vs impurity:
- Bernoulli loss $\sim$ entropy splitting
- Brier score $\sim$ gini splitting

RISK MINIMIZATION AND IMPURITY

- Tree fitting: Find best way to split parent node $\mathcal{N}_0$ into child nodes $\mathcal{N}_1$ and $\mathcal{N}_2$ such that $\mathcal{N}_1 \cup \mathcal{N}_2 = \mathcal{N}_0$ and $\mathcal{N}_1 \cap \mathcal{N}_2 = \emptyset$
- Two options for evaluating how good a split is: Per node $\mathcal{N}$ compute the following:
 - 1 Compute impurity $\text{Imp}(\mathcal{N})$ directly from observations in $\mathcal{N}$
 - 2 Fit optimal constant using loss function, sum up losses for $\mathcal{N}$
- Two common impurity measures are entropy and Gini index where $\pi_k^{(\mathcal{N})}$ are predicted probs for class $k = 1, \dots, g$ in node $\mathcal{N}$:

$$\text{Imp}^{\text{ent}}(\mathcal{N}) = - \sum_{k=1}^g \pi_k^{(\mathcal{N})} \log \pi_k^{(\mathcal{N})}$$

$$\text{Imp}^{\text{Gini}}(\mathcal{N}) = \sum_{k=1}^g \pi_k^{(\mathcal{N})} (1 - \pi_k^{(\mathcal{N})})$$

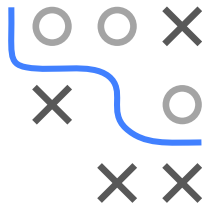


RISK MINIMIZATION AND IMPURITY

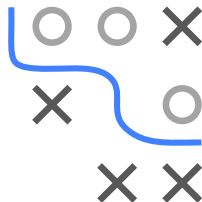
- In the following we will prove that entropy and Gini impurity measures are equivalent to splitting using log loss and Brier score:

$$L(y, \pi(\mathbf{x})) = - \sum_{k=1}^g \mathbb{I}[y = k] \log(\pi_k(\mathbf{x})) \quad (\text{log-loss})$$

$$L(y, \pi(\mathbf{x})) = \sum_{k=1}^g (\mathbb{I}[y = k] - \pi_k(\mathbf{x}))^2 \quad (\text{Brier})$$



RISK MINIMIZATION AND IMPURITY



$$\begin{aligned}\bar{\mathcal{R}}_{\text{emp}}(\mathcal{N}) &= \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x}, y) \in \mathcal{N}} \left(- \sum_{k=1}^g \mathbb{I}[y = k] \log \pi_k(\mathbf{x}) \right) \\ &= \frac{1}{n_{\mathcal{N}}} - \sum_{k=1}^g \sum_{(\mathbf{x}, y) \in \mathcal{N}} \mathbb{I}[y = k] \log \pi_k^{(\mathcal{N})} \\ &= \frac{1}{n_{\mathcal{N}}} - \sum_{k=1}^g n_{\mathcal{N}, k} \log \pi_k^{(\mathcal{N})} = \frac{1}{n_{\mathcal{N}}} - \sum_{k=1}^g (n_{\mathcal{N}} \cdot \pi_k^{(\mathcal{N})}) \log \pi_k^{(\mathcal{N})} \\ &= - \frac{1}{n_{\mathcal{N}}} n_{\mathcal{N}} \sum_{k=1}^g \pi_k^{(\mathcal{N})} \log \pi_k^{(\mathcal{N})} = \text{Imp}^{\text{ent}}(\mathcal{N})\end{aligned}$$

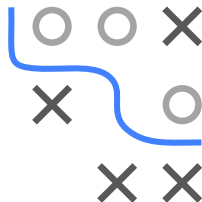
Avg. Bernoulli-risk of node $\mathcal{N}$ is equal to $\text{Imp}^{\text{ent}}(\mathcal{N})$

BRIER SCORE MINIMIZATION = GINI SPLITTING

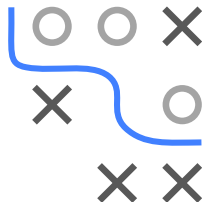
Inserting the optimal constant, the risk simplifies to

$$\begin{aligned}\bar{\mathcal{R}}_{\text{emp}}(\mathcal{N}) &= \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x}, y) \in \mathcal{N}} \sum_{k=1}^g (\mathbb{I}[y = k] - \pi_k^{(\mathcal{N})})^2 = \frac{1}{n_{\mathcal{N}}} \sum_{k=1}^g \sum_{(\mathbf{x}, y) \in \mathcal{N}} (\mathbb{I}[y = k] - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2 \\ &= \frac{1}{n_{\mathcal{N}}} \sum_{k=1}^g \left(\sum_{(\mathbf{x}, y) \in \mathcal{N}: y=k} (1 - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2 + \sum_{(\mathbf{x}, y) \in \mathcal{N}: y \neq k} (0 - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2 \right) \\ &= \frac{1}{n_{\mathcal{N}}} \sum_{k=1}^g n_{\mathcal{N},k} (1 - \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2 + (n_{\mathcal{N}} - n_{\mathcal{N},k}) (\frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}})^2\end{aligned}$$

since for $n_{\mathcal{N},k}$ observations the condition $y = k$ is met, and for the remaining $(n_{\mathcal{N}} - n_{\mathcal{N},k})$ observations it is not.



BRIER SCORE MINIMIZATION = GINI SPLITTING



We further simplify the expression to

$$\begin{aligned}\bar{\mathcal{R}}_{\text{emp}}(\mathcal{N}) &= \frac{1}{n_{\mathcal{N}}} \sum_{k=1}^g n_{\mathcal{N},k} \left(\frac{n_{\mathcal{N}} - n_{\mathcal{N},k}}{n_{\mathcal{N}}} \right)^2 + (n_{\mathcal{N}} - n_{\mathcal{N},k}) \left(\frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}} \right)^2 \\ &= \frac{1}{n_{\mathcal{N}}} \sum_{k=1}^g \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}} \frac{n_{\mathcal{N}} - n_{\mathcal{N},k}}{n_{\mathcal{N}}} (n_{\mathcal{N}} - n_{\mathcal{N},k} + n_{\mathcal{N},k}) \\ &= \frac{n_{\mathcal{N}}}{n_{\mathcal{N}}} \sum_{k=1}^g \pi_k^{(\mathcal{N})} \cdot (1 - \pi_k^{(\mathcal{N})}) = \text{Imp}^{\text{Gini}}(\mathcal{N})\end{aligned}$$

Avg. Brier-risk $\bar{\mathcal{R}}_{\text{emp}}(\mathcal{N})$ of the node is equal to its gini-impurity $\text{Imp}^{\text{Gini}}(\mathcal{N})$

WEIGHTING FOR RISK AND IMPURITY

- The emp. risk of a *split* is given by sum of per-node risks
($n_0 = n_1 + n_2$ are number of obs in nodes):

$$\begin{aligned}\mathcal{R}_{\text{emp}}(\text{split}) &= \mathcal{R}_{\text{emp}}(\mathcal{N}_1) + \mathcal{R}_{\text{emp}}(\mathcal{N}_2) = n_1 \bar{\mathcal{R}}_{\text{emp}}(\mathcal{N}_1) + n_2 \bar{\mathcal{R}}_{\text{emp}}(\mathcal{N}_2) \\ &= n_1 \text{Imp}(\mathcal{N}_1) + n_2 \text{Imp}(\mathcal{N}_2) \\ &= n_0 \left(\frac{n_1}{n_0} \text{Imp}(\mathcal{N}_1) + \frac{n_2}{n_0} \text{Imp}(\mathcal{N}_2) \right)\end{aligned}$$

- As you can see above: if working with average risk, we need to reweight in the addition (as the averages are computed on subsets of potentially unequal sizes)
- Average risks are used in impurity formulas, we so simply have to adhere to that slight modification in split finding with them
- The impurity of a split is defined as a weighted average

$$\text{Imp}(\text{split}) = \frac{n_1}{n_0} \text{Imp}(\mathcal{N}_1) + \frac{n_2}{n_0} \text{Imp}(\mathcal{N}_2)$$

